

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (EE) Examination

Dec 2012

EE 707 Advanced Power System Dynamics and Stability

Date: 11.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Explain the different methods for prevention of voltage stability. 07

OR

Q - 1 Derive the complete mathematical model of separately excited DC exciter and draw its block diagram.

Q - 2 A. Draw the PV and QV curve for voltage stability analysis and explain the procedure to draw the curves. 07

B. Discuss the V - Q sensitivity analysis method for voltage stability. 07

OR

A. Explain the modeling requirements for voltage stability analysis 07

B. Discuss the factors which affect the dynamic performance of excitation system. 07

Q - 3 Attempt any two 14

(a) Derive the block diagram of single machine infinite bus system with classical generator model.

(b) Explain the following w.r.t. excitation system (1) excitation system stabilizing circuit and (2) terminal voltage transducer and load compensator

(c) Discuss the construction and operating principle of static synchronous series compensator (SSSC)

SECTION - II

Q - 4 Derive the equations for turbine-generator torsional characteristics 07

OR

Q - 4 List out all types of AC excitation system and explain any one in detail with complete block diagram. 07

Q - 5 A. Discuss the sub synchronous resonance phenomena in power system in details 07

B. Explain the V - I characteristic of static var compensator 07

OR

Q - 5 A. Discuss the analytical methods for SSR analysis 07

- B. Explain the transition of the system state during system disturbance for the improvement of system stability 07

Q – 6

Attempt any two

14

- (a) Power system stabilizer
- (b) How STATCOM is better as compared to SVC?
- (c) Advantages of TCSC on dynamic performance of power system
